

Think Before You Loop: "Big O Thinking" for PowerShell devs

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Many thanks to our sponsors:

Devolutions



Jane Street



Microsoft



SynEdgy



Jonas Sommer Nielsen

- PowerShell Summit 2015
- 2021 KMD - Cloud Architect
- Aarhus, Denmark

Think. Measure. Profile.



Thinking is free.



Measuring is cheap.



Profiling is for when your already in trouble.

So, what is BIG O?

Big O

$O(n)$

worst case
Time or Space
complexity

N = biggest number

M = Other numbers?

1 = Constants

We are looking for the worst case, so throw away everything but the biggest “number”

Every loop inside a loop multiplies.



If you spot it before you write it, you save hours.



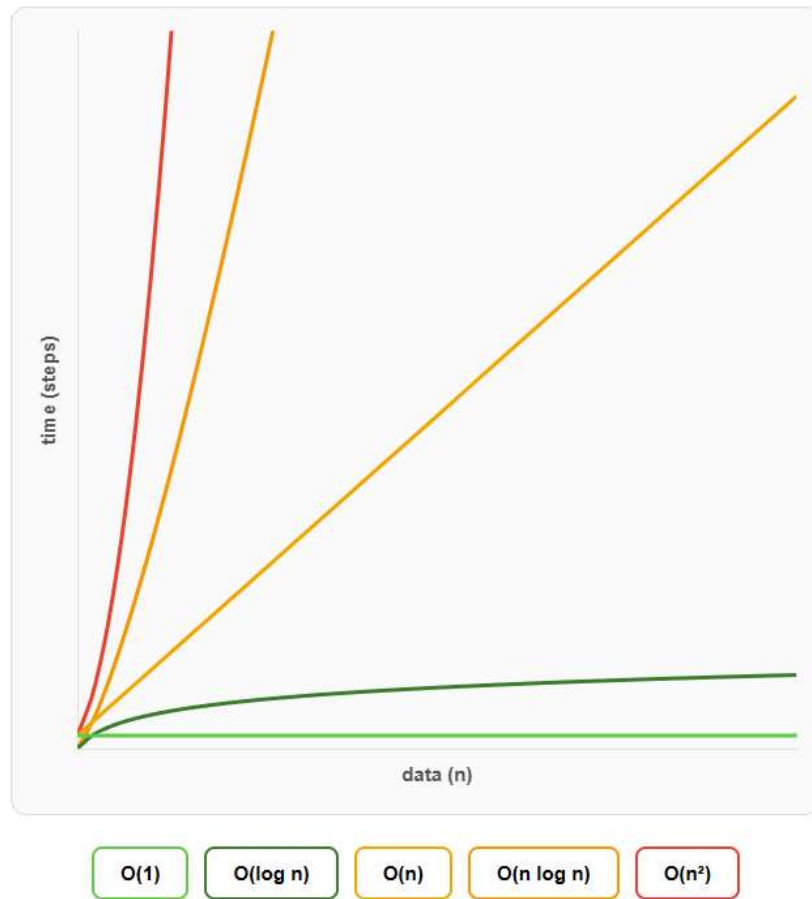
If you spot it after, you reach for Measure-Command or the Profiler.

Demos

We love Demos!



Big O notation



- $O(1)$
- $O(\log n)$
- $O(n)$
- $O(n \log n)$
- $O(n^2)$

What does Log n mean?

Binary Search Tree — $O(\log n)$

"Double the data, take just one more step."

NODES (N)

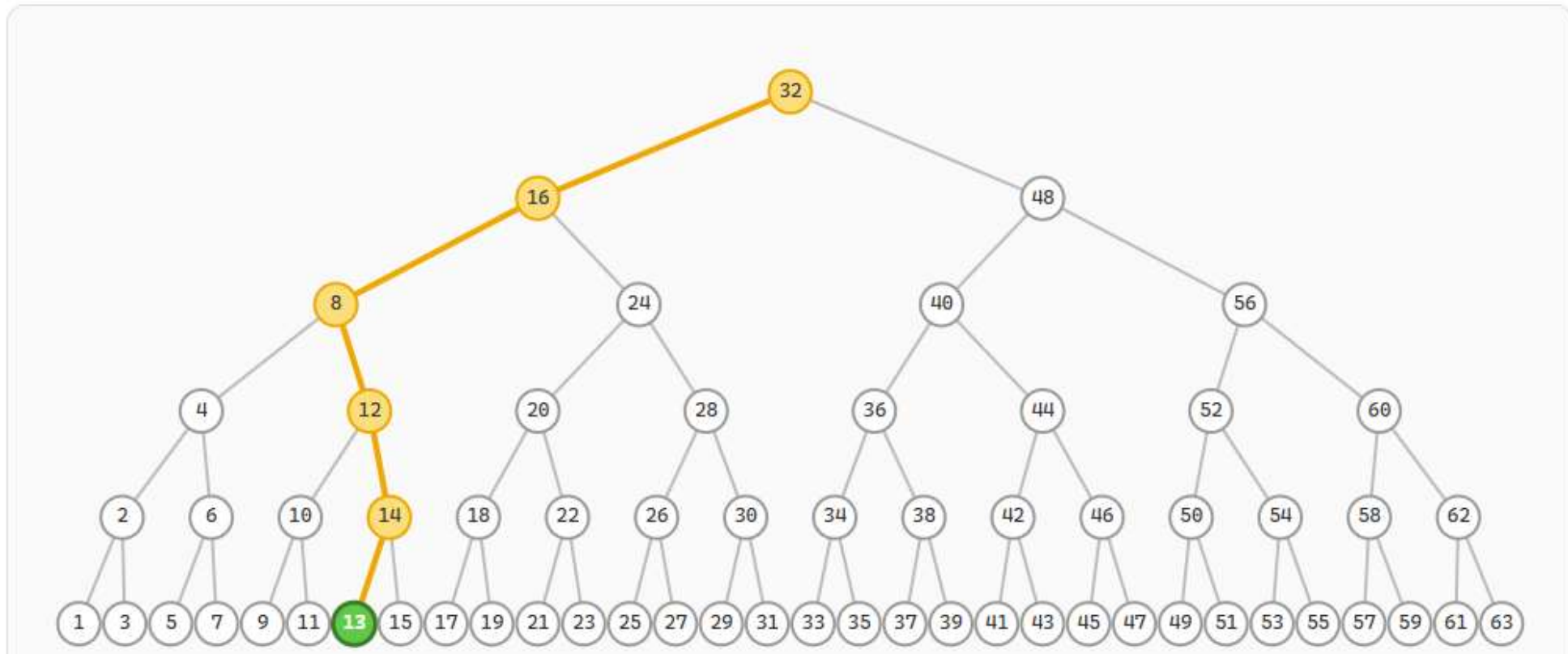
63

DEPTH (STEPS TO FIND ANY ITEM)

6

$\text{LOG}_2(N) + 1$

6



PowerShell scripting performance considerations

<https://learn.microsoft.com/en-us/powershell/scripting/dev-cross-plat/performance/script-authoring-considerations>

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Q&A

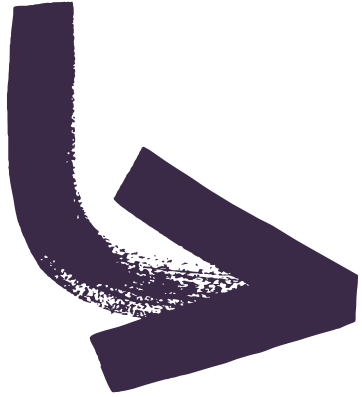
15 minutes

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